

SPM sampling-error band: delta method vs bootstrap

msamodel dev validation

Contents

1	What this document validates, and why it exists	1
2	The quantities	1
3	Delta method (the shipped formula)	2
4	Bootstrap (the independent reference)	2
5	Results — site axis	3
6	Results — mode axis	6
7	Per-quantity delta $\leftrightarrow$ bootstrap correspondence	9
8	Conclusion	9

1 What this document validates, and why it exists

The `predict*_ml` band functions report a **finite-mutation sampling error** (the “SPM arm”) in addition to the (`a1`, `a2`) parameter-uncertainty arm. The SPM arm is computed by a delta-method formula. This document is the durable, reproducible check that that formula is correct: it recomputes every SPM band from the **shipped package internals** and compares it, per residue, to an **independent bootstrap** — a genuinely different computation (resampling), so it can disagree when the delta formula is wrong.

Knit with

```
rmarkdown::render("dev/reports/spm_band_validation.Rmd")
```

2 The quantities

Let M be the SPM divergence matrix `dr2_ijm` (rows k = mutants, columns i = response sites). For selection strengths (a_1, a_2) the model assigns each mutant an unnormalised fixation probability p_k (`prefix_msa`), normalised to weights $w_k = p_k / \sum_k p_k$. The model’s predicted per-site divergence is the self-normalised weighted mean, and the fit works in its log and mean-centred forms:

$$\text{msd}_i = \sum_k w_k M_{ki}, \quad \text{lrmsd}_i = \frac{1}{2} \log \text{msd}_i, \quad \text{nlrmsd}_i = \text{lrmsd}_i - \frac{1}{|S|} \sum_{s \in S} \text{lrmsd}_s, \quad (1)$$

the last being the mean-centred quantity (over the response support S) that the fit works on. The four nested variants MM/MS/MA/MSA are the same formulas at $(0, 0)/(a_1, 0)/(0, a_2)/(a_1, a_2)$; the decomposition contributions are the contrasts $\phi_{\text{mut}} = \text{nlrmsd}_{\text{mm}}$, $\phi_{\text{stab}} = \text{nlrmsd}_{\text{ms}} - \text{nlrmsd}_{\text{mm}}$, $\phi_{\text{act}} = \text{nlrmsd}_{\text{msa}} - \text{nlrmsd}_{\text{ms}}$.

Only the finite mutant sample is random here (parameters are held fixed); the band is the sampling standard deviation of these quantities.

3 Delta method (the shipped formula)

The package computes a **per-mutant contribution matrix** h (internal `spm_hmat`). With $G_{ki} = M_{ki}/(2 \text{msd}_i)$ the log-sensitivity (the derivative of $\frac{1}{2} \log \text{msd}$):

$$G_{ki} = \frac{M_{ki}}{2 \text{msd}_i}, \quad (\text{if centred: } G_{ki} \leftarrow G_{ki} - \frac{1}{|S|} \sum_s G_{ks}) \quad (2)$$

$$h_{ki} = w_k \left(G_{ki} - \sum_{k'} w_{k'} G_{k'i} \right). \quad (3)$$

The centring subtraction $G - \text{rowMeans}(G)$ is $\partial \text{nlrmsd} / \partial M$: each mutant drives the support mean too. Then

- **Profile band** (single model): $\text{SE}(\text{lrmsd}_i) = \sqrt{\sum_k h_{ki}^2}$.
- **Contrast band** ($\phi = \text{lrmsd}_A - \text{lrmsd}_B$ on the *same* scan): $\text{SE} = \sqrt{\sum_k (h_{ki}^A - h_{ki}^B)^2}$.

The contrast's shared-sample cross term is automatic: the same mutant k enters both h^A and h^B , so $\sum_k (h^A - h^B)^2 = \sum_k h_A^2 + \sum_k h_B^2 - 2 \sum_k h_A h_B$, the last being $-2 \widehat{\text{Cov}}$. Equivalently, for the uncentred single model the formula reduces to the recorded

$$\text{SE}(\text{lrmsd}_i) = \frac{1}{2 \text{msd}_i} \sqrt{\sum_k w_k^2 (M_{ki} - \text{msd}_i)^2}, \quad (4)$$

since the w -weighted mean of $G_{\cdot i}$ is exactly $\frac{1}{2}$.

$\text{Var} = \text{SE}^2$ is what the predictors add to the parameter-arm variance (under independence) before forming the band.

4 Bootstrap (the independent reference)

This is the reference the delta formula is checked against — the correct bootstrap for this problem, not a simplified one. Per replicate: draw N mutant rows **uniformly with replacement** — this is how the scan drew them; the model weights are *not* sampling probabilities. Drawing with probability proportional to w would double-count the weights (once in the draw, once in the mean) and inflate the profile SD by about 10% for weighted ensembles (it vanishes only for uniform MM). On the drawn multiset, recompute each model's Hájek weighted mean

$$\widetilde{\text{msd}}_i^{(v)} = \frac{\sum_{k \in \text{draw}} w_k^{(v)} M_{ki}}{\sum_{k \in \text{draw}} w_k^{(v)}}, \quad (5)$$

take $\frac{1}{2} \log$, centre, and difference (a **shared** draw across variants preserves the cross-model correlation for ϕ). The band is the SD across B replicates.

5 Results — site axis

Table 1: Site axis: per-quantity agreement. Ratio near 1 across all 11 quantities.

quantity	median delta/boot	q10	q90	min delta SD
MM (uncentred)	1.008	0.987	1.026	0.037
MM (centred)	1.007	0.987	1.028	0.036
MS (uncentred)	1.009	0.986	1.029	0.030
MS (centred)	1.009	0.987	1.032	0.027
MA (uncentred)	1.006	0.987	1.030	0.019
MA (centred)	1.006	0.986	1.030	0.016
MSA (uncentred)	1.006	0.984	1.030	0.020
MSA (centred)	1.006	0.987	1.031	0.017
phi_mut (centred)	1.007	0.987	1.028	0.036
phi_stab (centred)	1.010	0.986	1.037	0.016
phi_act (centred)	1.001	0.971	1.030	0.008

Effective sample sizes $N_{\text{eff}} = (\sum p)^2 / \sum p^2$: MSA=1182, MS=1608, MA=1688, MM=2280 (N=2280). The weighted ensembles use about 28% of the mutants effectively, so the between-model correlation is large — which is why the ϕ contrast **must** keep the -2Cov cross term (a variance-sum formula is off by roughly 3x here).

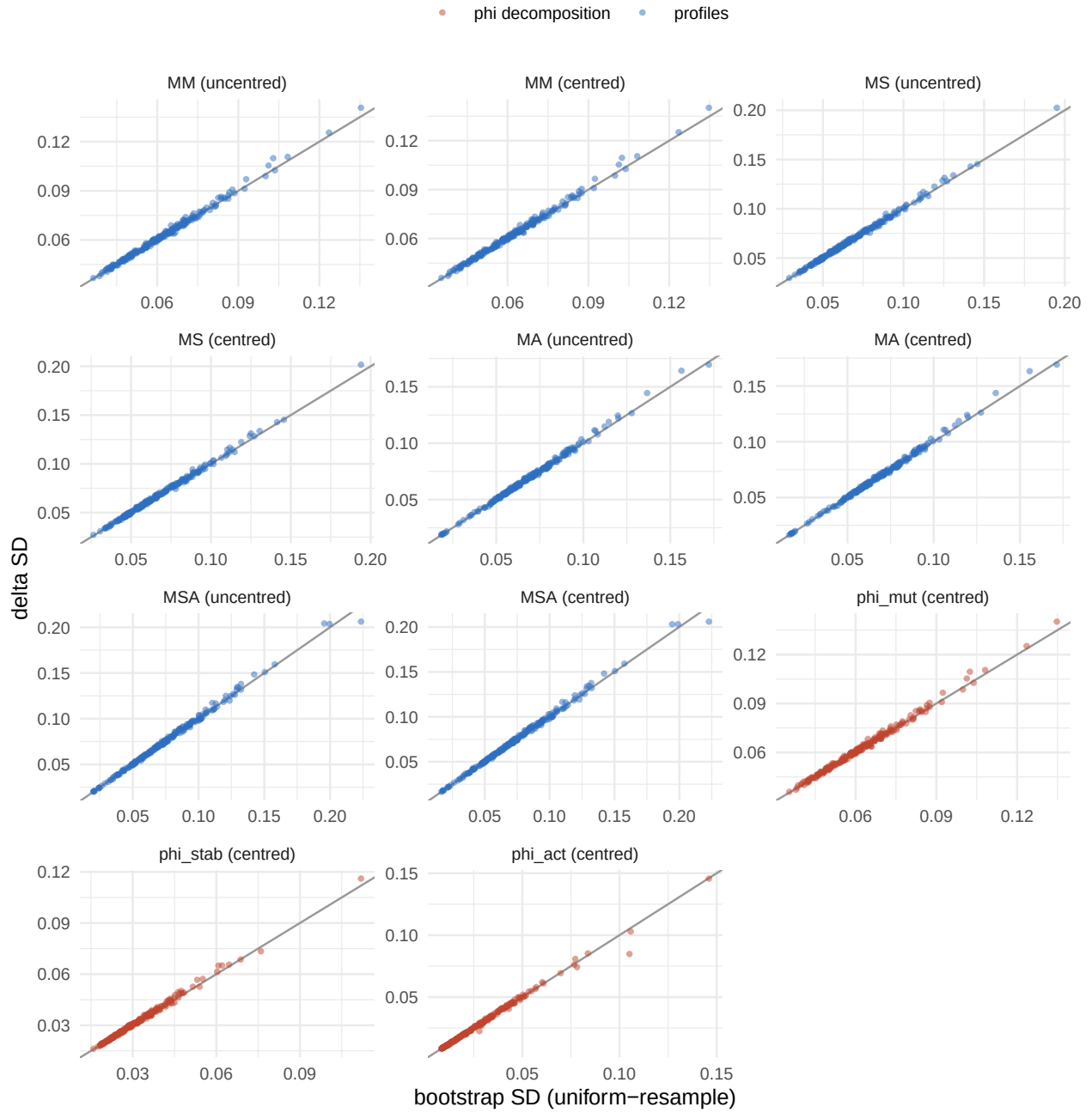


Figure 1: Site axis: per-site delta SD vs bootstrap SD. Points on the $y=x$ line = agreement.

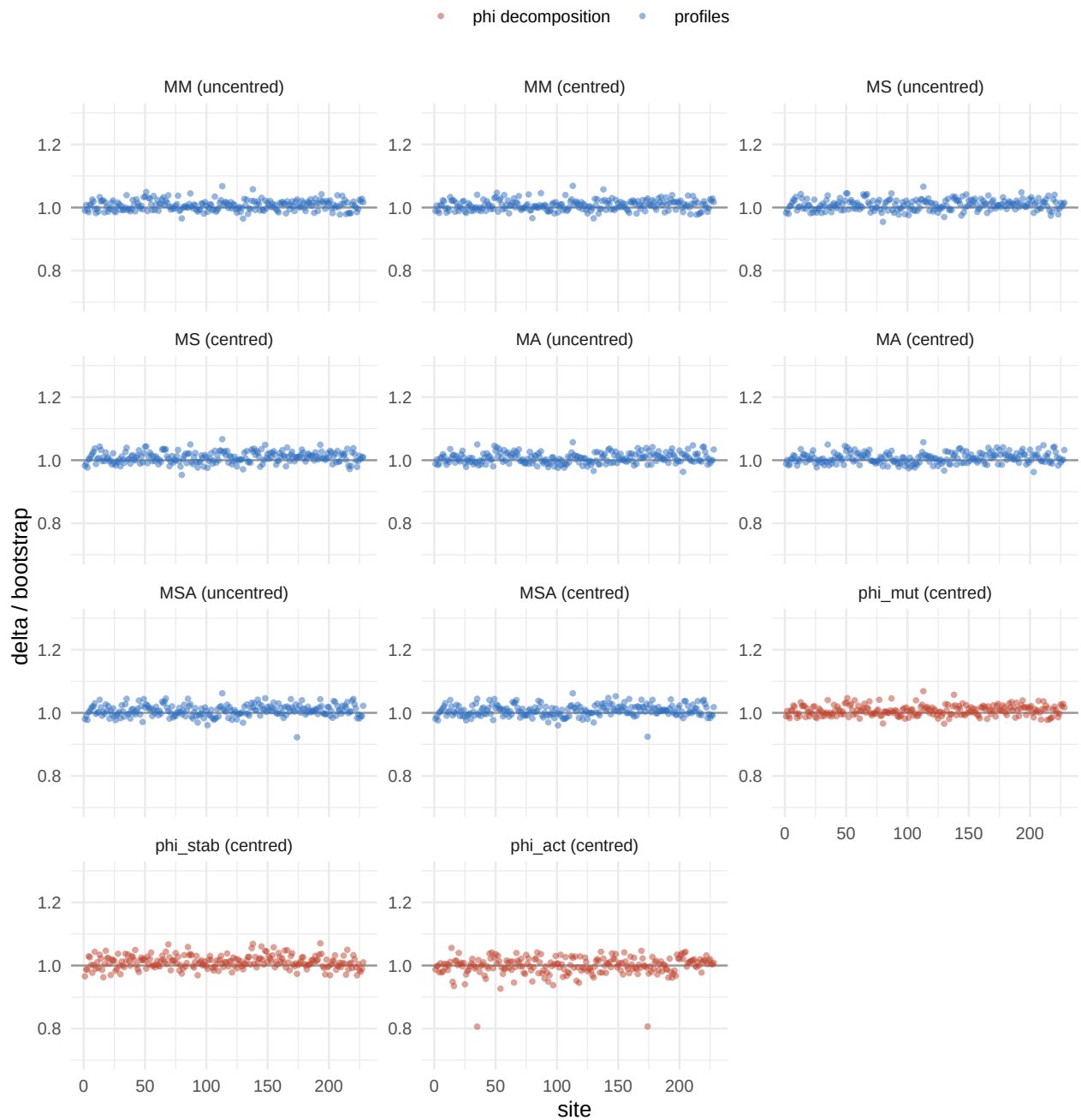


Figure 2: Site axis: per-site ratio $\text{delta} / \text{bootstrap}$ (1.0 = agreement).

6 Results — mode axis

The mode axis is the exact parallel of the site axis: the same delta and bootstrap procedures, on the per-mode divergence matrix `dr2_njm` (from `preprocess_spm_mode`) instead of the per-site `dr2_ijm`, and the mode fit `fit_lrmsd_n_msa_ml`. It is validated separately because the two axes have separate preprocessing and response matrices.

Table 2: Mode axis: per-quantity agreement. Ratio near 1 across all 11 quantities.

quantity	median delta/boot	q10	q90	min delta SD
MM (uncentred)	1.000	0.982	1.022	0.017
MM (centred)	1.000	0.981	1.021	0.017
MS (uncentred)	1.000	0.981	1.022	0.019
MS (centred)	1.000	0.981	1.021	0.018
MA (uncentred)	1.000	0.981	1.022	0.020
MA (centred)	0.999	0.979	1.022	0.019
MSA (uncentred)	1.000	0.980	1.023	0.021
MSA (centred)	1.000	0.979	1.023	0.020
phi_mut (centred)	1.000	0.981	1.021	0.017
phi_stab (centred)	1.003	0.983	1.026	0.009
phi_act (centred)	1.001	0.976	1.024	0.006

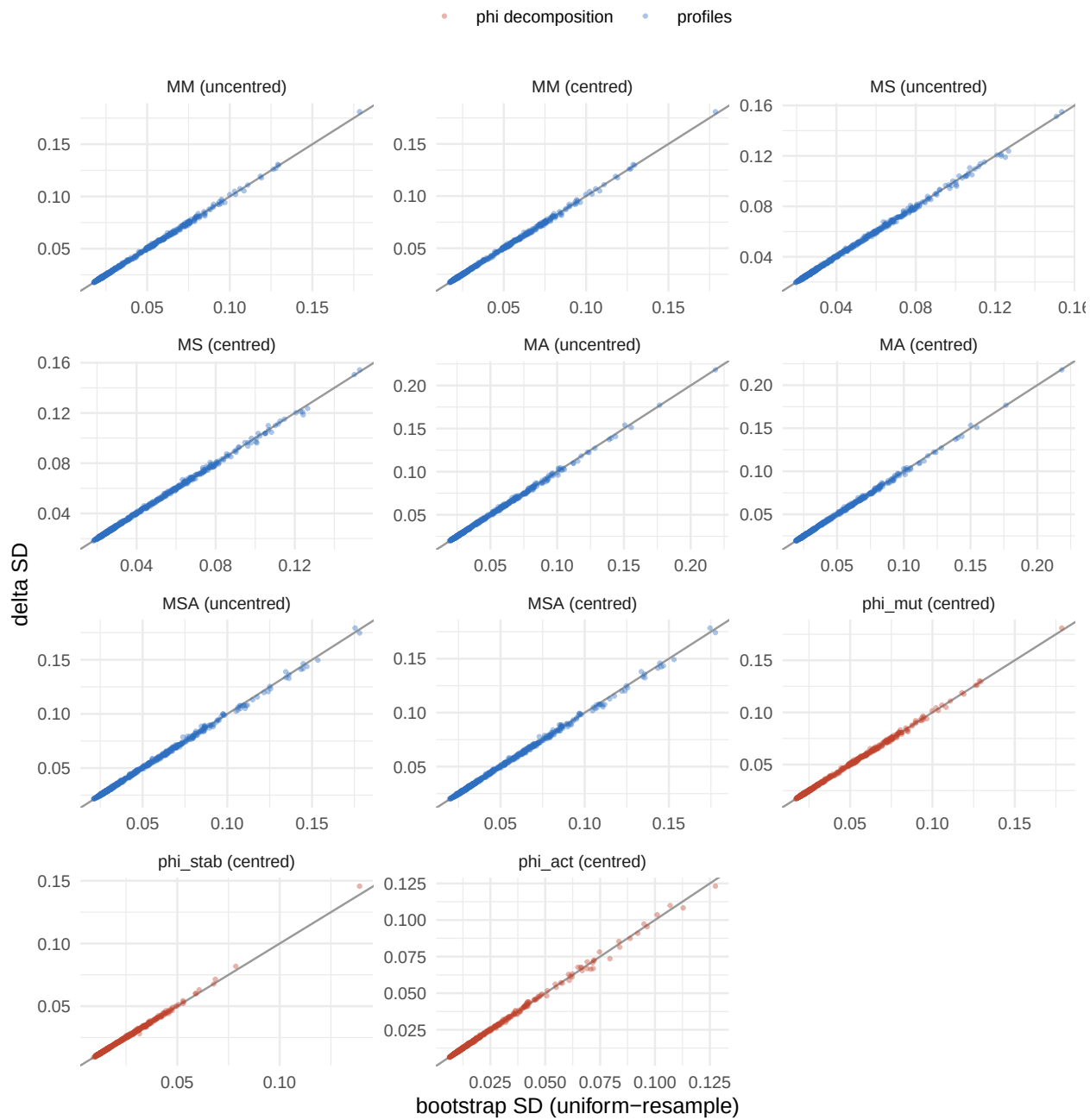


Figure 3: Mode axis: per-mode delta SD vs bootstrap SD.

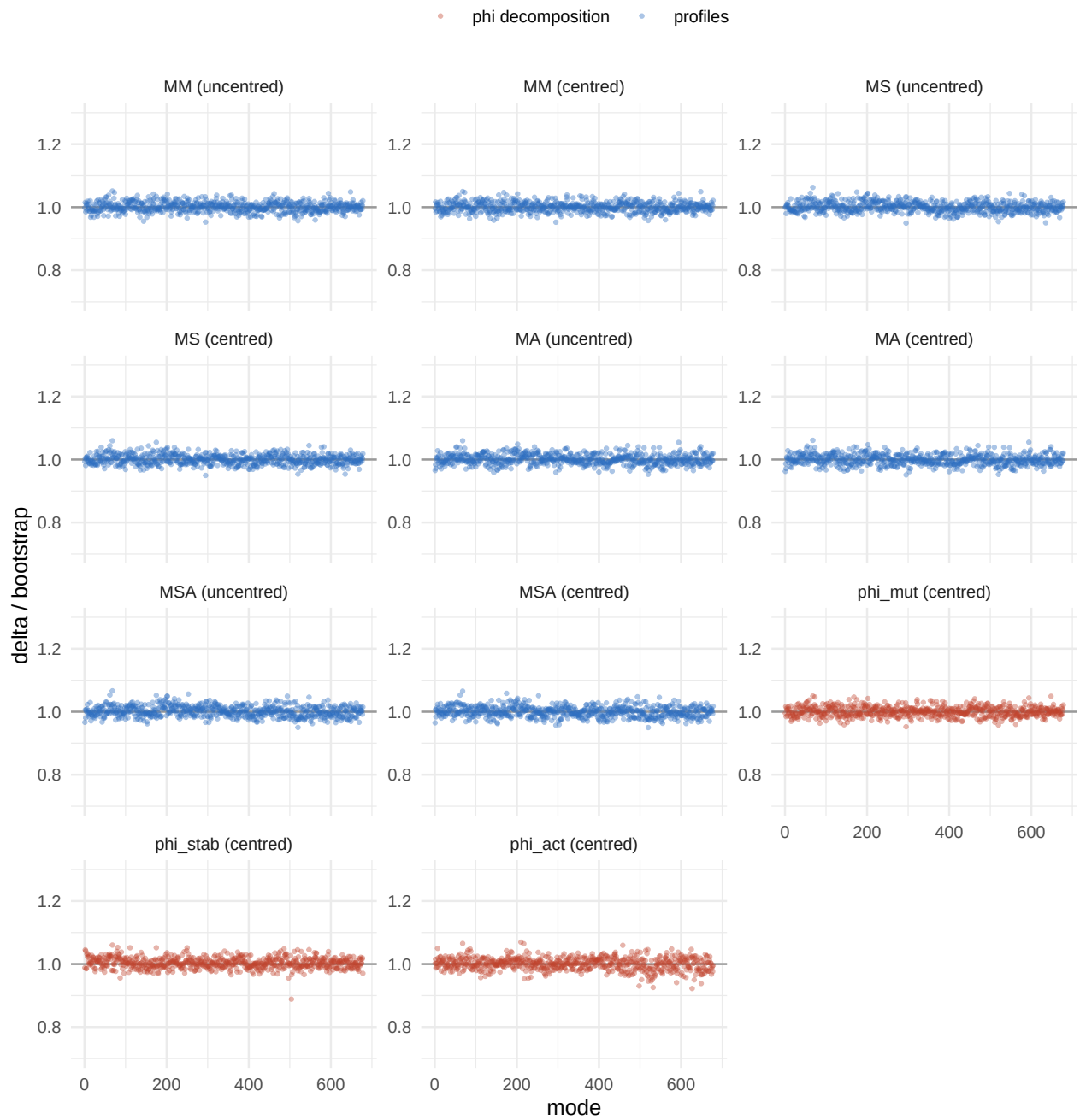


Figure 4: Mode axis: per-mode ratio $\text{delta} / \text{bootstrap}$ (1.0 = agreement).

7 Per-quantity delta $\leftrightarrow$ bootstrap correspondence

Table 3: What each quantity’s delta formula and bootstrap procedure are.

Quantity	Delta (shipped)	Bootstrap
MM/MS/MA/MSA (uncentred)	$\sqrt{\sum_k h^2}$, $h = \text{spm_hmat}(M, w, \text{FALSE})$	SD of $0.5 * \log(\text{Hajek msd_v})$, uniform draw
MM/MS/MA/MSA (centred)	$\sqrt{\sum_k h^2}$, $h = \text{spm_hmat}(M, w, \text{TRUE})$	SD of $(0.5 * \log \text{msd_v} - \text{mean})$, same draw
phi_mut	= nlrmsd_mm band (single MM model)	SD of nlrmsd_mm draws
phi_stab	$\sqrt{\sum_k (h_{\text{ms}} - h_{\text{mm}})^2}$, shared scan	SD of $(\text{nlrmsd_ms} - \text{nlrmsd_mm})$, shared draw
phi_act	$\sqrt{\sum_k (h_{\text{msa}} - h_{\text{ms}})^2}$, shared scan	SD of $(\text{nlrmsd_msa} - \text{nlrmsd_ms})$, shared draw

8 Conclusion

On **both** the site axis and the mode axis, all eleven quantities — the four nested variants in uncentred and centred form, and the three decomposition contributions — agree between the shipped delta formula and the independent bootstrap to within Monte-Carlo noise. The delta SPM band is validated.